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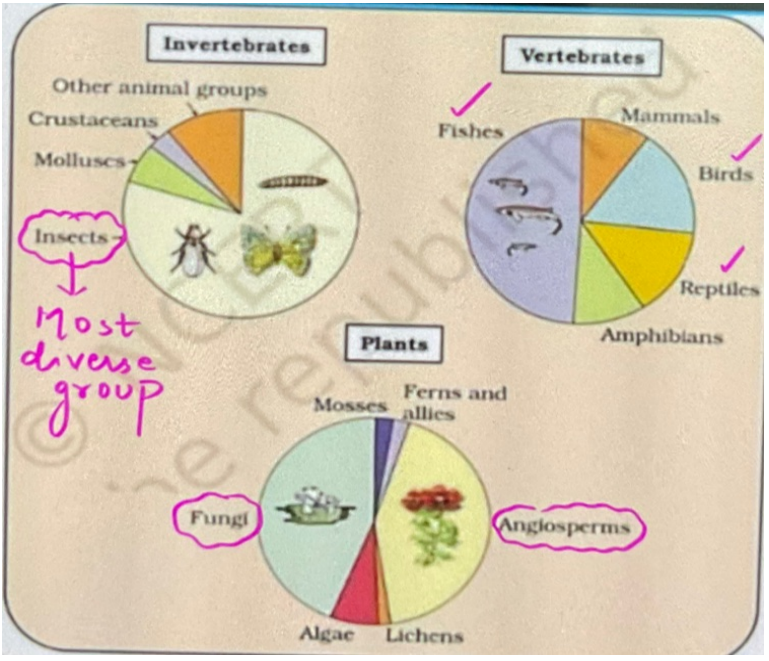
Biodiversity and Conservation

The most important of them are–

(i) **Genetic diversity**: A single species might show high diversity at the genetic level over its distributional range. The genetic variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges might be in terms of the potency and concentration of the active chemical (reserpine) that the plant produces. India has more than 50,000 genetically different strains of rice, and 1,000 varieties of mango.

(ii) **Species diversity**: The diversity at the species level. For example, the Western Ghats have a greater amphibian species diversity than the Eastern Ghats.

(iii) **Ecological diversity**: At the ecosystem level. India, for instance, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows has a greater ecosystem diversity than a Scandinavian country like Norway.



15.1.2 Patterns of Biodiversity

(i) ***Latitudinal gradients:*** The diversity of plants and animals is not uniform throughout the world but shows a rather uneven distribution. For many group of animals or plants, there are interesting patterns in diversity, the most well- known being the latitudinal gradient in diversity. In general, species diversity decreases as we move away from the equator towards the poles. With very few exceptions, tropics (latitudinal range of 23.5° N to 23.5° S) harbour more species than temperate or polar areas. Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds. A forest in a tropical region like Equador has up to 10 times as many species of vascular plants as a forest of equal area in a temperate region like the Midwest of the USA. The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth- it is home to more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and of more than 1,25,000 invertebrates. Scientists estimate that in these rain forests there might be at least two million insect species waiting to be discovered and named.

polar

What is so special about tropics that might account for their greater biological diversity? Ecologists and evolutionary biologists have proposed various hypotheses; some important ones are (a) Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long

Alexander Von Humboldt

Rectangular hyperbola

$S = CA^Z$

$\log S = \log C + Z \log A$

$Y = C + mx$

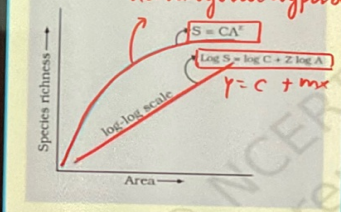


Figure 15.2 Showing species area relationship. Note that on log scale the relationship becomes linear

$\log S = \log C + Z \log A$

where

S = Species richness ✓ A = Area ✓

Z = slope of the line (regression coefficient) ✓

C = Y-intercept ✓

ving species area relationship. that on log scale the relationship nes linear

Ecologists have discovered that the value of Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or the region (whether it is the plants in Britain, birds in California or molluscs in New York state, the slopes of the regression line are amazingly similar). But, if you analyse the species-area relationships among very large areas like the entire continents, you will find that the slope of the line to be much steeper (Z values in the range of 0.6 to 1.2). For example, for frugivorous (fruit-eating) birds and mammals in the tropical forests of different continents, the slope is found to be 1.15. *What do steeper slopes mean in this context?*

0.6 - 1.2

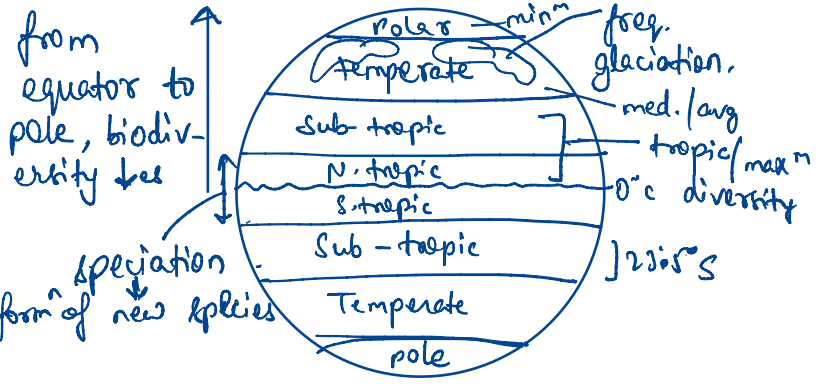
Species = Birds
= Plants
= Molloscs

steepest

Patterns of diversity

↗ longitudinal gradient

↘ species area - relationship.



evolutionary time for species diversification, (b) Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity and (c) There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

The Importance of Species Diversity to the Ecosystem

- ✓ Number of species in a community really matter to the functioning of the ecosystem because rich biodiversity is important for:
  - Stability
  - Productivity
  - Resilience
  - Alternative pathways
  - Health of ecosystems

a) Stability:

- Communities with more species tend to be more stable than those with less species
  - i. A stable community should not show too much variation in productivity from year to year.
  - ii. It must be resistant or resilient to occasional natural as well as man-made disturbances.
  - iii. It must be resistant to invasions by alien species.

Note: David Tilman's

- Long-term ecosystem experiments using outdoor plots confirmed that these attributes are linked to species richness in a community.
- He found that
  - a. Plots with more species showed less year-to-year variation in total biomass.
  - b. Increased diversity contributed to higher productivity.

Now it is clear that species diversity is important for productivity, stability and resilience.

b) Ecosystem Health:

↳ Rivet-Popper hypothesis

Airplane	Ecosystem
Rivets	Species
Rivets on the wings	Key species Keystone species

Loss of Biodiversity

The biological wealth of our planet has been declining rapid

Complete disappearance of extinction of species result in complete loss of genetic information contained in it

Natural extinction

Mass extinction

It is the dying off or extermination of a large number of species due to catastrophes

During the long period, since the origin and diversification of life on earth, there were five episode of mass extinction of species

Anthropogenic extinction

Caused directly or indirectly by human activities like habitat destruction over exploiting hunting pollution

Colonisation of tropical Pacific Islands by humans have resulted in extinction of more than 2000 species of native bird

The IUCN red list 2004 document the extinction of 784 species in the last 500 years

It includes

338 vertebrate

359. Invertebrates

87 plants

Recent extinction includes

The Dodo (Mauritius)

Quagga, (Africa)

Thylacine, (Australia)

Steller's sea cow (Russia)

And three subspecies (Bali, Java, Caspian) of Tiger

Some groups like amphibians appear to be more vulnerable to extinction. Presently some 15,500 species worldwide are facing the threat of extension. They include

- 12% of all bird species
- 23% of all mammals Species
- 32% of all amphibian species
- 31% of all gymnosperm species

Results of loss of Biodiversity

Loss of biodiversity in a region may lead to

- ✓ Decline in plant production or productivity.
- ✓ Reduced resistance to environmental perturbations like drought.
- ✓ Increased variability of ecosystem processes like productivity, water use, pest and disease cycles.

Causes of biodiversity losses

4 causes (evil quartet)

1. Habitat loss and fragmentation (most imp.)
2. Over exploitation
3. Alien species
4. Co-extinction

Susceptibility to extinction

Species more susceptible to extinction have following the population characteristics.

- ✓ Large body size (e.g., Rhinoceros, Lion)
- ✓ Small population size and low reproductive rate (e.g., Giant Panda, Blue Whale).
- ✓ High trophic level in food chain (e.g., Bald Eagle, Bengal Tiger).
- ✓ Fixed habitat and migratory routes (e.g., Whooping Crane, Blue Whale).

Why should we conserve Biodiversity?

Narrowly Utilization (Direct or economic Uses)

a) Food
b) Firewood
c) Fibre
d) Construction material
e) Industrial Products
f) Drugs.

More than 25% of the drugs currently sold in the market worldwide are derived from a mere 120 species of plants About 25,000 species of plants contribute to the traditional medicines used by native peoples around the world

Broadly Utilitarian (ecosystem services)

a) Oxygen
b) Pollination
c) Aesthetic Pleasure
d) flood and erosion control

Ethical / moral values.

How Do We Conserve Biodiversity?

Two basic strategies of biodiversity conservation:

a) *in-situ* (on site)

b) *ex-situ* (off site)

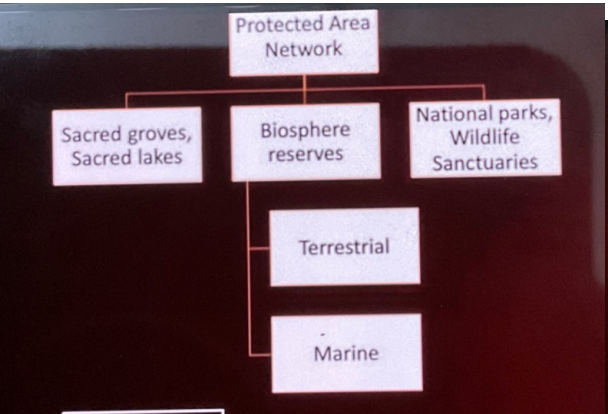
i. *In-situ* Conservation:

- Emphasise protection of whole ecosystem
- In-situ conservation strategies are of two types
  - a. Hot spots and
  - b. Protected areas

a. **Hot spots:**

- Richest and the most threatened reservoirs of plant and animal life on earth.
- The key criteria for determining a hot spot are:
  1) Very high levels of species richness
  2) High degree of endemism
  3) Degree of threat, which is measured in terms of habitat loss
- Initially, 25 hot spots were identified globally
- Now the number is raised to 34 occupying earth's land area of less than 2%
- Strict protection of these hot spots could reduce the ongoing mass extinctions by almost 30%
- India has three hotspots:
  1) Western Ghats and Sri Lanka
  2) Indo-Burma
  3) Himalaya

b. Protected areas:



a) **National Parks:**

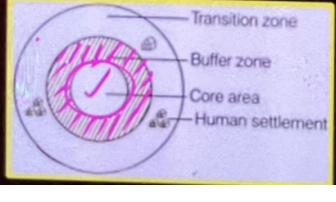
- Maintained by government
- Reserved for betterment of wildlife (flora and fauna both)
- Grazing, felling of trees, habitat manipulation and cultivation are not allowed
- The Jim Corbett National Park was the first National Park established in India

b) **Sanctuaries:**

- These are tracts of land where animals (fauna) are protected from all types of exploitation and habitat disturbance.
- Collection of forest products, harvesting of timber, tilling of land, private ownership etc. are allowed.

c) **Biosphere Reserves**

- Tribal people are an integral component of the system
- The concept of Biosphere Reserves was launched in 1975 as a part of UNESCO's Man and Biosphere Programme (MAB)
- 14 biosphere reserves in India



d) Sacred Groves:

- These are found in several parts of India:
  1. Khasi and Jaintia Hills in Meghalaya
  2. Aravalli Hills of Rajasthan
  3. Western Ghat regions of Karnataka and Maharashtra
  4. Sarguja, Chanda and Bastar areas of Madhya Pradesh
- In Meghalaya, the sacred groves (sacred forest) are the last refuges for a large number of rare and threatened plants.

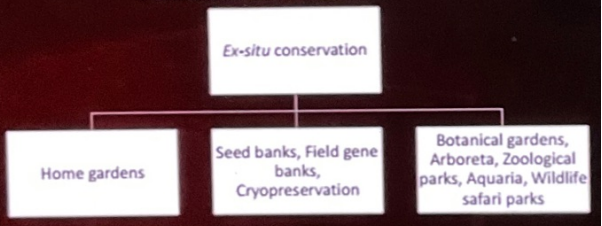
The main functions of biosphere reserves are:

1. Conservation
2. Development
3. Scientific research, monitoring and education

World summit ↓ 2002,

ii. ***Ex-situ* Conservation:**

- Threatened animals and plants are kept out from their natural habitat.



Conventions on Biodiversity

a) The Earth Summit

- This is the historic convention on Biological diversity held in Rio de Janeiro in 1992.
- It is called upon all the nations to take appropriate measures for:
  i. Conservation of biodiversity and
  ii. Sustainable utilization of the benefits from biodiversity

THANK You !!